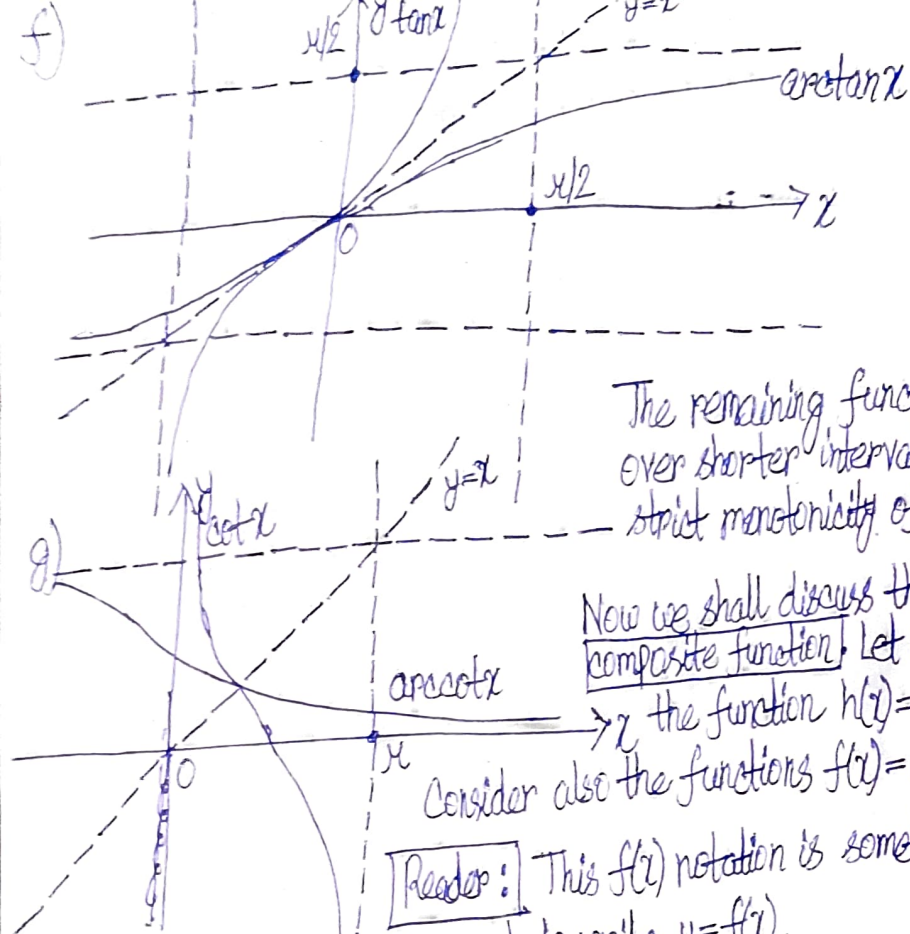




The remaining functions are defined over shorter intervals to ensure the strict monotonicity of the functions.

Now we shall discuss the concept of a composite function. Let us take an example the function $h(x) = \sqrt{1 + \cos^2 x}$.

Consider also the functions $f(x) = \cos x$ and $g(y) = \sqrt{1 + y^2}$.

Reader: This $f(x)$ notation is something new. So far we used to write $y = f(x)$.

Author: You are right. However, it is expedient to simplify the notation. Consider the three functions: $h(x)$, $f(x)$ and $g(y)$.

The function $h(x)$ is a composite function composed of $f(x)$ and $g(y)$: $h(x) = g[f(x)]$.

Reader: I understand. Here the values of $f(x)$ are used as the values of the independent variable (argument) for $g(y)$.

Author: Let us have a look at Fig. 21, which pictures the mappings of sets in the case of our composite function, $h(x) = \sqrt{1 + \cos^2 x}$, with $f(x) = \cos x$ defined over the interval $[0, \pi]$.

We see that the function f is a mapping of D (the interval $[0, \pi]$) onto G (the interval $[-1, 1]$), that is, the mapping f . The function g (the function

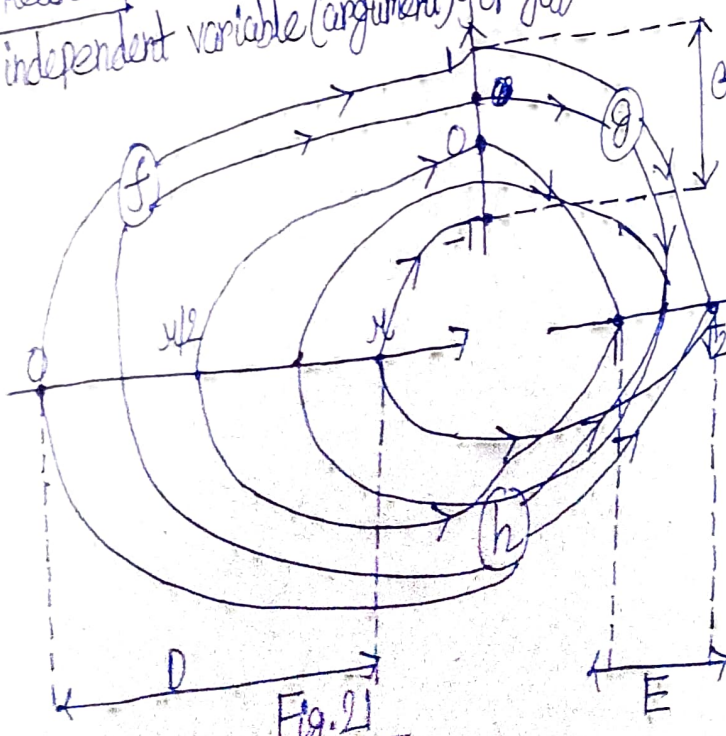


Fig. 21